Description

Image





Caption

1. French horn. © Copper Development Association 2. Pump parts. © Copper Development Association

The material

Brasses are alloys of copper (Cu) with 5 - 40% zinc (Zn). They combine excellent formability and machinability with good corrosion resistance. As the zinc content increases the melting point, density, modulus and electrical and thermal conductivities decrease; the expansion coefficient, the strength and the hardness increase.

Compositional summary

Alpha,"70-30" or Cartridge brass: Cu + 30% Zn
Alpha-Beta, "60-40" or Naval brass: Cu + 40% Zn, often with a little lead

General properties

Density		8.07e3	-	8.48e3	kg/m^3
Price	*	5.4	-	5.56	GBP/kg
Date first used		-1200			

Mechanical properties

Young's modulus		95	-	110	GPa
Shear modulus	*	35	-	40	GPa
Bulk modulus	*	100	-	110	GPa
Poisson's ratio		0.34	-	0.35	
Yield strength (elastic limit)		100	-	296	MPa
Tensile strength		280	-	511	MPa
Compressive strength	*	100	-	296	MPa
Elongation		15	-	52	% strain
Hardness - Vickers		60	-	150	HV

Fatigue strength at 10^7 cycles	*	140	-	207	MPa
Fracture toughness	*	42.6	-	76	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	6.7e-5	-	3.4e-4	

Thermal properties

Melting point		887	-	997	°C
Maximum service temperature		127	-	193	°C
Minimum service temperature		-236			°C
Thermal conductor or insulator?		Good conductor			
Thermal conductivity		99.4	-	138	W/m.°C
Specific heat capacity	*	374	-	382	J/kg.°C
Thermal expansion coefficient		18.2	-	20.6	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good conductor			
Electrical resistivity		5.9	-	8.4	µohm.cm

Optical properties

Transparency		Opaque			

Critical Materials Risk

High critical material risk?		Yes			

Processability

Castability		4	-	5	
Formability		4	-	5	
Machinability		4	-	5	
Weldability		4	-	5	
Solder/brazability		5			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Acceptable			
Soils, alkaline (clay)		Excellent			
Wine		Acceptable			

Durability: acids

Acetic acid (10%)		Limited use			
Acetic acid (glacial)		Unacceptable			

Citric acid (10%)	Limited use
Hydrochloric acid (10%)	Unacceptable
Hydrochloric acid (36%)	Unacceptable
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Unacceptable
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Limited use
Phosphoric acid (85%)	Unacceptable
Sulfuric acid (10%)	Limited use
Sulfuric acid (70%)	Unacceptable

Durability: alkalis

Sodium hydroxide (10%)	Excellent
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Limited use
Benzene	Excellent
Carbon tetrachloride	Excellent
Chloroform	Excellent
Crude oil	Acceptable
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Limited use
Toluene	Excellent
Turpentine	Excellent
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Limited use
Acetone	Excellent
Ethyl alcohol (ethanol)	Excellent
Ethylene glycol	Acceptable
Formaldehyde (40%)	Acceptable
Glycerol	Excellent
Methyl alcohol (methanol)	Excellent

Durability: halogens and gases

Chlorine gas (dry)	Excellent
Fluorine (gas)	Acceptable
O2 (oxygen gas)	Limited use
Sulfur dioxide (gas)	Limited use

Durability: built environments

Industrial atmosphere	Acceptable
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Excellent

Durability: flammability

Flammability	Non-flammable

Durability: thermal environments

Tolerance to cryogenic temperatures	Excellent
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Acceptable
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.85e7			tonne/yr
Reserves, principal component	6.9e8			tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 4.38	-	4.82	kg/kg
Embodied energy (virgin grade production)	* 62.5	-	68.9	MJ/kg
Water usage	* 306	-	338	l/kg

Material processing: energy

Metal casting, embodied energy	* 41.1	-	46	MJ/kg
Metal extrusion, hot, embodied energy	* 8.8	-	10	MJ/kg
Metal rolling and forging, embodied energy	* 4.87	-	5.54	MJ/kg
Wire drawing, embodied energy	* 13.1	-	14.5	MJ/kg
Metal powder forming, embodied energy	* 35.3	-	39.8	MJ/kg
Chemical vapor deposition (CVD), embodied energy	* 8.15e3	-	9.83e3	MJ/kg

Machining, coarse, embodied energy	* 15.8	-	17.5	MJ/kg
Machining, fine, embodied energy	* 18.2	-	20.2	MJ/kg
Machining, grinding, embodied energy	* 20.6	-	22.9	MJ/kg
Machining, laser, CO2, embodied energy	* 0.97	-	1.17	MJ/m

Material processing: climate change

Metal casting, climate change (CO2-eq)	* 2.89	-	3.25	kg/kg
Metal extrusion, hot, climate change (CO2-eq)	* 0.576	-	0.658	kg/kg
Metal rolling and forging, climate change (CO2-eq)	* 0.318	-	0.363	kg/kg
Wire drawing, climate change (CO2-eq)	* 0.86	-	0.95	kg/kg
Metal powder forming, climate change (CO2-eq)	* 2.85	-	3.22	kg/kg
Chemical vapor deposition (CVD), climate change (CO2-eq)	* 671	-	808	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.03	-	1.14	kg/kg
Machining, fine, climate change (CO2-eq)	* 1.19	-	1.32	kg/kg
Machining, grinding, climate change (CO2-eq)	* 1.34	-	1.5	kg/kg
Machining, laser, CO2, climate change (CO2-eq)	* 0.0799	-	0.0962	kg/m

Recycling and end of life

Recycle	✓			
Functional recycle	✗			
Climate change (CO2-eq), recycling	* 0.964	-	1.06	kg/kg
Embodied energy, recycling	* 15	-	16.5	MJ/kg
Recycle fraction in current supply	26			%
Combust for energy recovery	✗			
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Brass is non-toxic and readily recycled

Supporting information

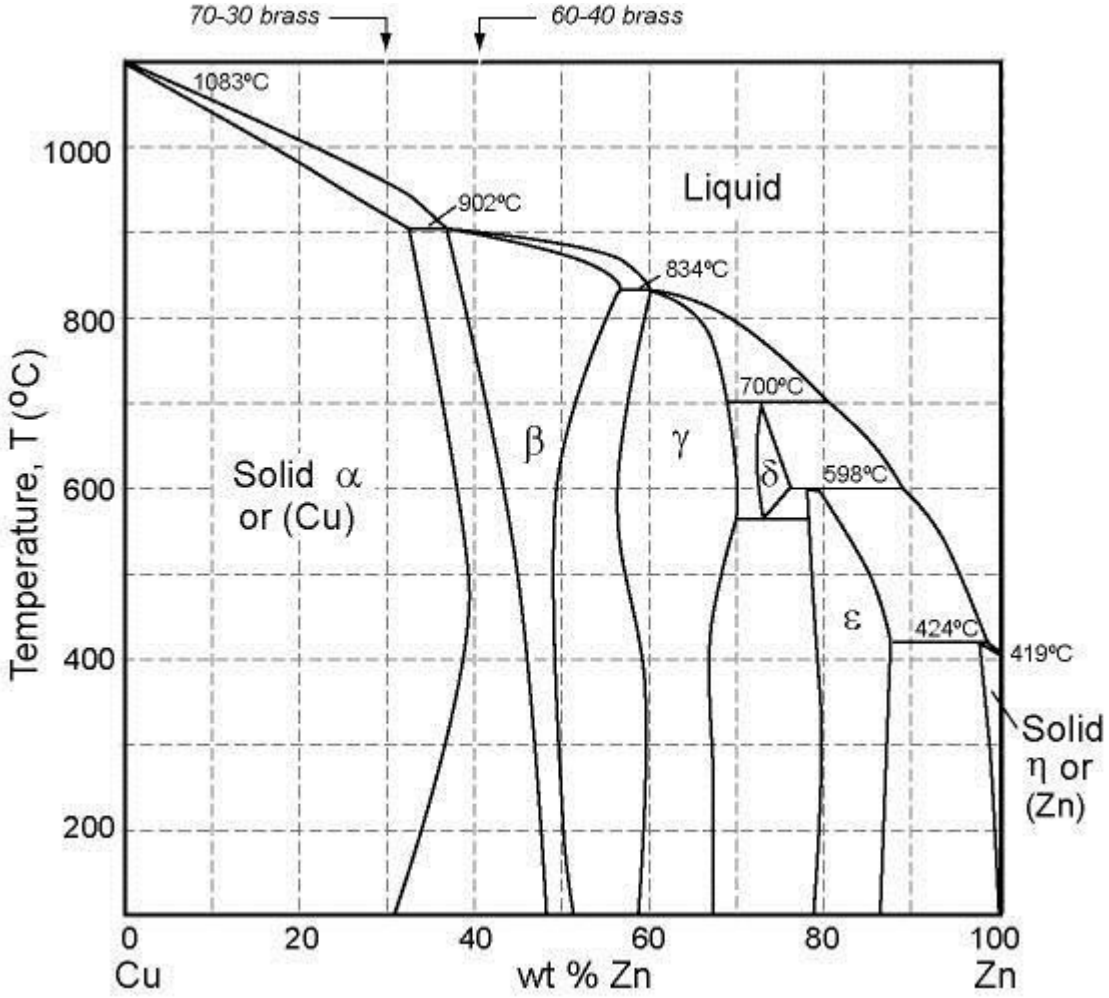
Design guidelines

Brasses are readily worked by casting, drawing, spinning and machining. They have a pleasing color ranging from the red of copper through browner bronze-like hues and gold (jewelry brass closely resembles the color of 14 carat gold).

Technical notes

There is now a UNS designation system for copper and its alloys: the letter C (for 'copper') followed by a 5-digit number. Only the first digit means anything: C1**** designates almost pure copper, the C2, C3 and C4 series are brasses with increasing zinc content, the C5s are bronzes based on copper and tin, the C6s are other bronzes containing aluminum instead of tin, and the C7s are copper-nickel alloys.

Phase diagram



Phase diagram description

Brasses are alloys of copper (Cu) with 5 - 40% zinc (Zn), for which this is the phase diagram. The two most widely used compositions are 70-30 brass and 60-40 brass. Their compositions are shown.

Typical uses

70-30 Brasses: deep drawn parts such as cartridge and shell casings, musical instruments and other objects made by sheet-metal working, decorative components for architecture, marine components, condenser and cooling units, jewelry.60-40 Brasses: tubing for condensers, heat exchangers, valve and pump parts, taps and water pipes, fasteners and springs.

Tradenames

Red brass (15% Zn); Jewelry brass (12.5% Zn); Yellow brass (35% Zn); Muntz metal (40% Zn).

Values marked * are estimates.
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Producers

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